The ECBench ECDLP Performance Frontier

End-to-end boundary, implementation census, bottlenecks, and the next isolated campaign

ECBench frontier audit

31 October 2026

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1 Executive answer

The repository does **not** have one global ECDLP performance frontier. It has several frontiers indexed by curve family, target law, advice model, accounting unit, and evidence class. Once those domains are kept separate, the boundary is clear:

1. **Generic prime, random-binary, and Koblitz elliptic curves remain rho-led.** No admitted cold, one-target index-calculus result is below its strongest applicable matched generic reference. The closest current repository markers are 2.28× rho for inherited F4 on one tiny random-binary cell, 2.59× for the prime residual collector at 36 bits, and 3.43× on the admitted Koblitz ladder. These are different studies and units, so they are boundary markers, not entries in one sortable table.
2. **One restricted curve family is IC-led.** The plane-quartic attack on the special GHS-weak family $E_0/\mathbb{F}_{q^3}$, with $\#E_0(\mathbb{F}_{q^3}) = 4\ell$, is below matched rho in its own cold, one-target comparison. The older B0 arm crosses below rho at $q = 4051$; the N4-v2 double-large-prime arm reaches 0.00523× negation-map rho at $q = 2{,}097{,}091$. This is a real restricted-family result. It is **not** a generic-curve, standard-curve, prime-order $E/\mathbb{F}_{q^3}$, or ECC2K-130 result.
3. **The best newly validated factor-base change is engineering progress, not a frontier crossing.** On the 20-bit prime holdout, the selected low-x base grows from $N = 32$ to $N = 45$ and costs 0.836, 0.817, and 0.746 of the incumbent at native-operation prices $w = 0, 0.1, 1$. It still costs 7.95, 8.24, and 9.62 times matched `rho.negation`. The result was reproduced on a second L2 EKS host, but covers one curve and gives no exponent.
4. **F5 and F6 improve local solver work without improving E2E ECDLP cost.** F5 uses 0.641–0.647 of matrix F4's word work at degree 13. F6 uses roughly 0.77–0.82 of inherited-F4 word work and shortens the relation phase by 1.7–1.9×, but its charged closure/addition work makes its whole-pipeline lower-bound cost 64.6–1,352.6× inherited F4 and 399–4,883× rho on the tested cells.
5. **The optimized rho comparison is materially stronger at the implementation level.** The trajectory-identical signed-Frobenius rho path uses 0.802155 of the same-binary legacy Callgrind instructions, a 1.246642× speedup, while counted GAE and all outcomes remain identical. Its native wall improvement remains unpromoted because no L2 wall session completed. The new campaign therefore uses accounting-v2 rho, a same-target A/A rho control, and also measures BSGS and kangaroo where they can be finite-size leaders.

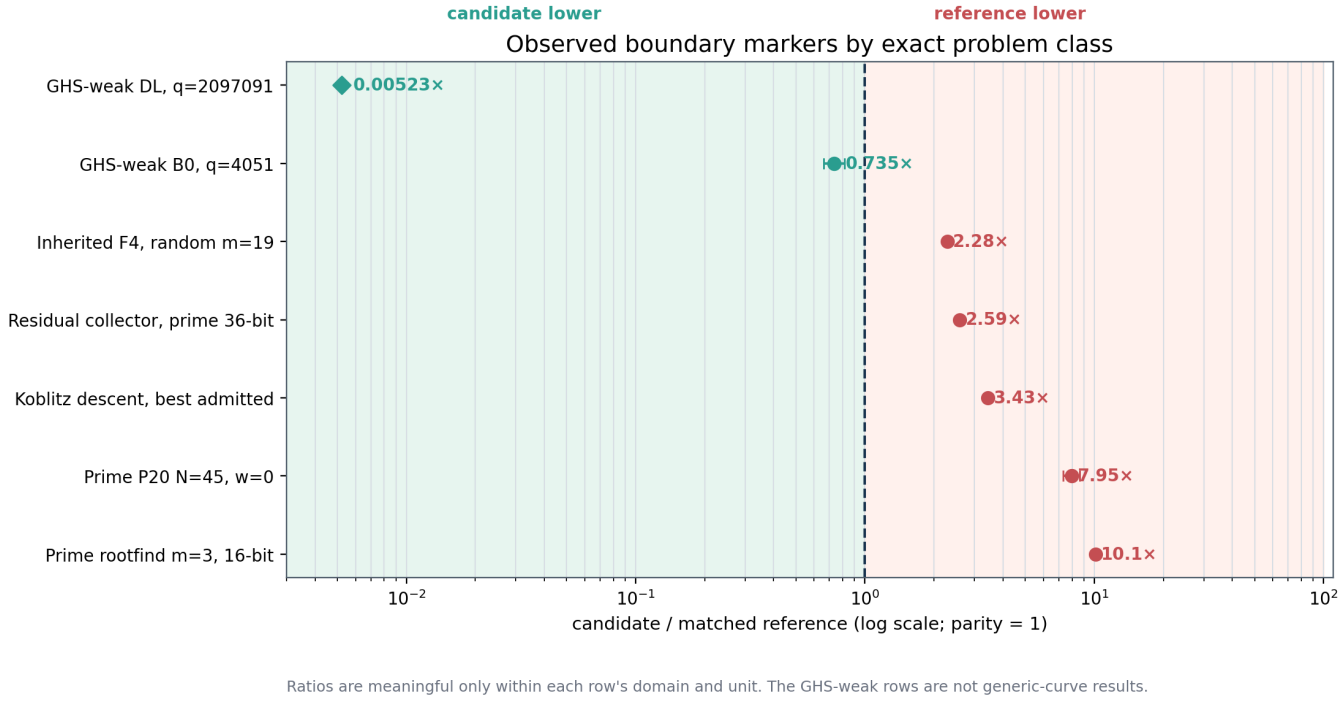


Figure 1: Boundary markers. A ratio is meaningful only inside its row's exact domain and unit.

This report is an evidence synthesis at source commit `48c8a4767d2d1eed6a7b379f74410bc729239380`. It does not present blocked jobs as measurements. The exact current-source EKS campaign is prepared, but no authenticated AWS/Iso Lab route is available in this workspace; consequently, there is no fresh 2026-10-31 wall-time leaderboard.

1.1 Boundary by domain

Domain	Current finite boundary	Leader	Evidence reading
Prime-field generic ECDLP	closest recent IC marker 2.59× matched rho; P20 validated base row 7.95–9.62×	generic algorithms	rho remains lower; BSGS can beat rho on several toy bitop cells
Random-binary generic ECDLP	inherited F4 2.28× rho on the tiny m=19 diagnostic; admitted ladder 10.6–38.2×	generic algorithms	no IC win
Koblitz generic ECDLP	admitted cold one-target IC 3.43–11.5× matched rho	signed-Frobenius generic walk	no IC win
Special GHS-weak $E_0/\mathbb{F}_{q^3}$	B0 0.735× at q=4051; N4-v2 DL 0.00523× negation rho at q=2,097,091	plane-quartic IC	restricted-family IC win
Hyperelliptic Jacobians	genus-4 toy row 0.114× rho in bitop/v1	Jacobian IC	not an elliptic-curve result
Standard cryptographic curves / ECC2K-130	no complete break or scaling evidence	generic security boundary	unresolved by experiment; no claim of a break

2 What "the frontier" formally means

2.1 Domain key

A benchmark row is assigned to the exact domain

$$D = (\mathcal{C}, G, \mathcal{T}, A, k, U, P, E),$$

where $\mathcal{C}$ is the curve family and representation, G is the subgroup and cofactor treatment, $\mathcal{T}$ is the target distribution, A is reusable advice or preprocessing, k is the number of targets charged together, U is the accounting unit, P is the native-operation pricing rule, and E is the evidence class. Two observations from different domain keys do not form a speed ratio. In particular:

- prime GAE, binary word XORs, SAT conflicts, FES Gray-code vector XORs, and native wall seconds are not interchangeable;
- a one-target cold run and a retained-setup online run are different objects;
- a transfer to a special weak curve family is not evidence about the original generic family unless an explicit, subgroup-preserving, costed map is part of the run;
- L0 shared-host wall time cannot displace L2 isolated wall evidence.

2.2 Cost vector and dominance

For an admitted observation x in a fixed domain, retain the vector

$$C(x) = (W_0, W_{0.1}, W_1, T_{L2}, M, A, S, F),$$

where W_w is counted work with each unpriced native operation charged at $w \in \{0, 0.1, 1\}$ GAE, T_{L2} is isolated wall time, M is peak memory, A is reusable advice, S is success/censoring, and F is the retained failure set. Observation (x) dominates (y) only if both have the same domain key, (x) is no worse on every required axis, and it is strictly better on at least one axis. The frontier is the non-dominated set.

Unknown quantities remain unknown. They are never inserted as zero. A smaller backend-local counter can move a component frontier without moving the ECDLP frontier.

2.3 Admission gate

The refreshed ranking requires all of the following:

- exact curve, subgroup, target, factor-base, binary, and algorithm identity;
- cold one-target cost as the primary object;
- strongest applicable rho reference with the same usable symmetry, plus a same-target rho A/A control;
- verified scalar recovery and replay of every measured record;
- at least eight public targets and at least five measured rounds;
- at least four sizes before fitting or describing scaling;
- failures, timeouts, work limits, and censored pairs retained;
- the $w = 0, 0.1, 1$ pricing band whenever native operations are not fully converted;
- L2 or better isolation for a wall-time claim;
- no cross-ranking of backend-native units.

The full rules and machine-readable design are in PROTOCOL.md and protocol.json. The protocol validator reports zero findings, but the document remains `draft` until the exact current-source ECR digest is bound and committed before execution.

3 Implementation census

The release binary exposes **74 top-level method IDs**:

Family	Count
Index calculus and algebraic/transfer candidates	31
Rho and transfer references	32
BSGS	7
Kangaroo	2
Pair-claw	2

The algebraic pipeline separately exposes **11 nested solver backends**: `f4-f2`, Buchberger, matrix F4, matrix F5, inherited F4, Crossbred, XL, SAT-CDCL, scalar FES, wide FES, and exhaustive search. These are not eleven additional top-level ECDLP methods; they are solver choices inside a complete pipeline.

Every ID and backend has an explicit disposition in COVERAGE.md and coverage.json: 13 current-primary methods, 10 diagnostic methods, 21 legacy controls, 6 pricing-sensitivity methods, and 24 domain-specialized methods. "Legacy" does not mean incorrect, and "blocked" does not mean slow; it records why the method is not in the fresh primary campaign.

The exact native registry is preserved in method-registry.json. This prevents "all implementations" from silently meaning only the arms that appeared in an older report.

4 Generic prime-field frontier

4.1 Generic algorithms are the finite-size leaders

On the 16–40-bit prime `bitop/v1` ladder, rho, BSGS, kangaroo, the pair-table IC rule, and the m=3 root-finding rule are measured on public synthetic prime-order curves. BSGS is below the current rho implementation on many toy cells (for example 0.558× at 16 bits), while rho approaches the better scaling boundary. The IC pair rule ranges from 6.23× to 78.3× rho; root-finding ranges from 10.1× to 69.7× through 28 bits.

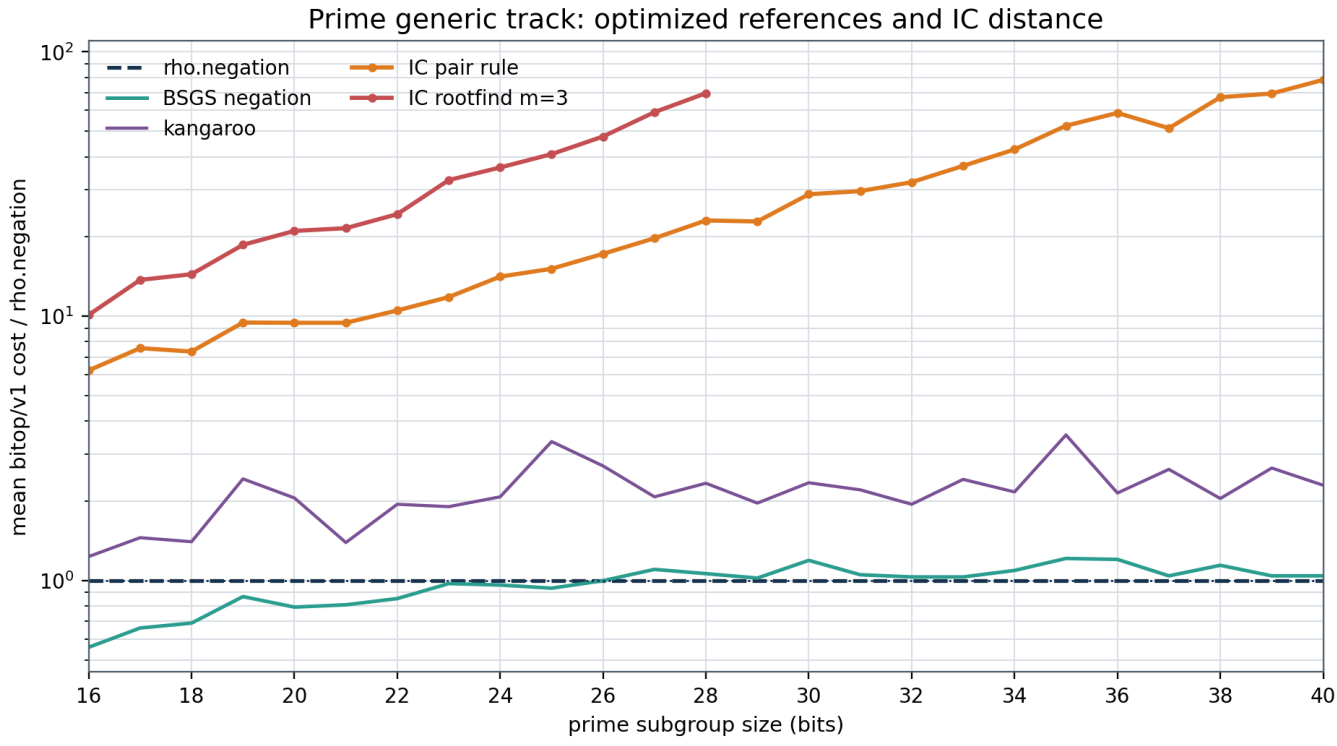


Figure 2: Prime finite-size ladder. BSGS is included so the rho reference is not mistaken for the absolute finite-size leader.

This is why the prepared campaign includes `bsgs.negation_v2` and `kangaroo.vow_v2` beside `rho.negation_v2`. Rho remains the asymptotic generic reference; it is not assumed to win every tiny finite instance.

4.2 Factor-base selection

The 2026-10-29 study jointly selected the low-x base size and exact pair oracle on training curves, then decided each size on a fresh holdout curve. The pair oracle stayed fixed; only (N) changed.

Bits	Incumbent → selected N	selected/incumbent at $w = 0/0.1/1$	selected/rho at $w = 0/0.1/1$	Decision
20	32 → 45	0.818 / 0.798 / 0.726 on frozen holdout; 0.836 / 0.817 / 0.746 on second host	7.95 / 8.24 / 9.62 on second host	promote N=45 engineering baseline
24	162 → 115	0.902 / 0.940 / 1.126	12.44 / 12.92 / 15.19	pricing relabelling; keep N=162
28	407 → 288	0.929 / 0.974 / 1.193	20.25 / 20.98 / 24.33	pricing relabelling; keep N=407
32	1024 → 1024	1 / 1 / 1	33.30 / 33.18 / 32.61	null selection

For P20, the second host admitted all 2,016 executions at L2, verified all 2,016, and matched 1,008/1,008 original records in every counted and band-priced field. The independent validation is filed in `VALIDATION.json`. The leaderboard's stale "no validation exists" sentence was corrected at the generator and generated Markdown/HTML/JSON surfaces.

The operational interpretation is narrow: increasing (N) helps the 20-bit pipeline because the improved relation probability pays back the larger base and table over that finite envelope. At 24 and 28 bits the decision changes with the native-operation price, so those candidates do not dominate the incumbent.

4.3 Root finding: useful algorithm, wrong asymptotic label if overclaimed

`rootfind-subtract` and the m=3 root-finding arm are exact generic table/search procedures. They can be optimized and ranked E2E, but they do not receive the factor-base equations inside an algebraic system. They therefore belong to the **generic decomposition track**, bounded by the square-root frontier, rather than serving as evidence for a sub-square-root algebraic attack. The repository's 16–28-bit measurements remain 10.1–69.7× rho in `bitop/v1`.

The older root-finder replay audit also records an open regression: only 72/168 current-binary replays were identical on 2026-10-19. Affected historical stage counters remain blocked from promotion. The fresh campaign measures the current implementation from new records rather than inheriting those counters.

4.4 Best recent prime marker and the missing algebraic baseline

The progress ledger's closest recent prime IC marker is the residual collector at 36 bits, 2.59× `rho.negation` in cold one-target (S). It is closer than the P20 factor-base study, but it is not a completed prime Gröbner-basis attack.

There is still no native prime-field ECBench arm that places the factor-base constraint inside a Semaev system and then runs a complete F4/F5/root-extraction pipeline against matched rho. Prime PKM/tower and GLV studies contain useful component counts, but their units and systems do not form that missing E2E baseline. This is the largest algorithmic coverage hole in the prime program.

4.5 Standalone GPU explicit relations

The RTX PRO 6000 study completes both fixed public 62-bit fixtures, with cold times 2,139.88 and 2,187.94 seconds in its V11 panel. Certificates replay and corruption controls pass. It remains **unranked** here because it is a standalone GPU implementation, its clocks are L0, and there is no same-fixture ECBench rho arm. It establishes an operational capability boundary, not a generic or matched-rho frontier.

5 Binary and Koblitz frontiers

5.1 Random binary

The admitted random-binary ladder remains 10.6–38.2× matched negation rho. In the newer F4/F5/F6 cross-family diagnostic, inherited F4 is closer on the tiny cells: 5.89× at m=13 and 2.28× at m=19. Two sizes do not establish an exponent, and the study's (S) is a lower bound because backend-native work is unpriced.

The key result is negative but useful: Koblitz-specific structure does not automatically improve the decomposer if the factor base is folded only by negation. The same solver ordering and similar local ratios appear on Koblitz and random-binary systems.

5.2 Koblitz

The admitted cold, one-target Koblitz table covers subgroup sizes around $(2^{\{36.6\})-(2}\{47.2\})$. IC/rho ranges from 3.43× to 11.5×. Setup accounts for roughly 98.7–99.6% of the cold IC pipeline on the central rows, so an online-only view can look attractive while the primary cold object remains rho-led.

The correct Koblitz reference uses signed Frobenius and negation. The canonical medium-tier counted boundary is 1.089× its floor, with interval [0.927, 1.261], across six sizes and 144 runs. Pair-claw is a useful memory/time frontier and reached 1.85× matched rho on one earlier cell, but did not displace the walk.

The factor-base symmetry lesson is asymmetric:

- folding a base or pair table can reduce stored representatives and setup;
- the matched rho reference must receive the same usable automorphism group;
- a symmetry that helps only a representation stage is not an ECDLP speedup;
- orbit naming, canonicalization, and retained advice must be charged to both arms consistently.

6 The restricted GHS-weak exception

The repository's only clear IC-led elliptic-curve domain is the special GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $\#E_0(\mathbb{F}_{q^3}) = 4\ell$. The comparison is a whole cold one-target pipeline in $\mathbb{F}_q$-multiplication equivalents, with no subgroup enumeration hidden outside the interval.

The B0 arm crosses below plain rho at $q = 4051$, where its ratio is 0.735 [0.662, 0.816] at $w = 0$, and falls to 0.122 [0.0987, 0.157] by $q = 262111$. N4-v2 adds double large primes, structured Gaussian elimination, and block Lanczos; its DL arm reaches 0.00456× plain rho and 0.00523× negation-map rho at $q = 2{,}097{,}091$ in the recorded counted unit.

This result must travel with four restrictions:

1. the curves are intentionally GHS-weak and are not generic curves;
2. they are not generic prime-order $E/\mathbb{F}_{q^3}$ instances;
3. no standard cryptographic curve is in this family;
4. the result cannot be transported to another curve without an explicit, subgroup-preserving, costed map.

The correct synthesis is therefore not "IC never beats rho" and not "ECDLP is broken." It is: **IC beats rho on one special weak family; generic ECC remains rho-led in the measured envelope.**

7 F4, F5, F6, SAT, WDSAT, and FES

7.1 F4/F5/F6 result

The cross-family experiment uses eight targets in each of K13, R13, K19, and R19. Matrix F4/F5 at m=19 hit the retained 1,800-second timeout. At m=13:

Cell	Matrix F4 word XORs	Matrix F5 word XORs	F5/F4	Inherited F4 word XORs	F6 word XORs
K13	2.245e9	1.454e9	0.6465	1.112e7	9.10e6
R13	2.325e9	1.495e9	0.6414	8.66e6	7.99e6
K19	timeout	timeout	unknown	6.70e8	5.19e8

Cell	Matrix F4 word XORs	Matrix F5 word XORs	F5/F4	Inherited F4 word XORs	F6 word XORs
R19	timeout	timeout	unknown	7.15e8	5.62e8

F5's signature criterion removes about a third of matrix F4's elimination work on the same tree. Inherited F4 is roughly 199–222× cheaper than matrix F4 on the m=13 systems. F6 further reduces word work and relation wall time, but performs exact factor-base point additions for its closure, which dominate the common priced pipeline.

The frontier rule follows directly: F5 and F6 are component leaders in their native scopes, while inherited F4 remains the best of these four in the common E2E lower-bound (S). None beats rho.

7.2 SAT-CDCL

SAT-CDCL is live in the nested registry and supports CNF-expanded parity and native-XOR parity. Existing n=17 records show why conflict counts cannot be a universal unit: native parity reports thousands of conflicts while CNF parity reports more than a million on related systems, with different clauses, auxiliaries, and encodings. A conflict reduction is a solver diagnostic unless CNF construction, XOR handling, relation yield, verification, and the whole pipeline share a common priced boundary.

The new solver-backend spec pins encoding and budget and runs SAT beside the same curve, factor base, target seeds, rho reference, and A/A control. Until that L2 session completes, SAT is implemented but not a ranked E2E leader.

7.3 WDSAT

WDSAT is a historical external path rather than a current native ECBench method ID. XG/Gaussian-elimination mode can cut its own conflict count by tens or hundreds of times, but "conflict" changes meaning with solver, encoding, and XG mode. More importantly, a retained audit found a false-UNSAT result on one of sixteen decomposable targets in the unfixed path. The historical WDSAT evidence is therefore useful for regression and solver research, not for promotion to the current E2E frontier.

7.4 FES and Gray code

`fes-f2` uses scalar Gray-code enumeration; `fes-f2-wide` evaluates multiple subcubes through AVX2/AVX-512 and is the strongest exhaustive reference where its equation limit applies. Gray-code word XORs and vector XORs are native units, not GAE. Past component studies show real implementation gains, but a universal gate was not cleared on every cell. The fresh all-backend spec keeps both implementations, their timeouts, and their native counters, while ranking only common E2E outcomes.

7.5 Crossbred, XL, Buchberger, and exhaustive controls

These backends remain important controls:

- Buchberger checks that matrix techniques are not winning by an incorrect algebraic shortcut;
- Crossbred tests hybrid specialization/elimination;
- XL provides a degree-lifted linearization baseline;
- exhaustive search supplies the correctness and small-system floor.

Their native counter definitions differ, so no report should divide one by another and call the quotient a speedup. The common comparison starts at the same ECDLP target and ends only after the recovered scalar verifies.

8 Rho comparison: current optimized boundary

The optimized `rho.signed_frobenius_strong` path preserves the exact walk, recovered scalar, counters, and GAE while reducing implementation instructions:

Measurement	Result
Optimized / same-binary legacy Callgrind instructions	0.802155

Measurement	Result
Optimized / archived pre-change instructions	0.804582
Speedup over same-binary legacy	1.246642×
Canonicalization share	53.91% → 42.48%
Verified executions	864 / 864
Measured replays identical	720 / 720
GAE optimized / legacy	1.0000 [1.0000, 1.0000]
Native L2 wall verdict	unknown

The implementation win comes from paired coordinate maps, fixed common byte shapes, and shared coefficient reduction. It does not change the mathematical rho operations boundary. The old L0 wall ratio is not promoted.

For the new campaign:

- prime uses `rho.negation_v2` and a byte-identical A/A arm;
- Koblitz uses `rho.signed_frobenius_strong_v2`, also with A/A;
- BSGS and kangaroo are measured on prime cells to reveal finite-size leaders;
- every symmetry available to an IC arm is audited for the reference;
- setup, distinguished-point policy, table state, restarts, and verification stay inside the declared interval.

9 Bottleneck map

Bottleneck	Evidence now	Why it holds the frontier	Required next gate
Relation probability / trials	prime and binary IC gaps grow across several ladders	solver improvements do not change the probability of a useful decomposition	algebraic factor-base constraint inside the system; four-size E2E sweep
Factor-base and pair-table setup	98.7–99.6% of central cold Koblitz rows; $O(\|F\|^2)$ pair materialization	online-only improvements disappear when one target pays setup	cold primary result; equal-advice online secondary result
F6 closure additions	64.6–1,352.6× inherited F4 in lower-bound S	local word-XOR savings are overwhelmed by exact group work	price closure and solver in one common unit; change closure algorithm, not label
Matrix F4/F5 scale	m=19 matrix arms time out at 1,800 s	symbolic preprocessing and matrix construction explode before useful scaling	memory/degree receipts and retained timeout curve
Solver-counter mismatch	XORs, conflicts, vector steps, multiply-adds	cross-solver ratios lack semantic equality	E2E wall or complete common pricing; native counters remain diagnostics
Linear algebra	large stage-only speedups, small E2E changes in several pipelines	relation collection and setup dominate	measure phase share before further LA work
Rho reference quality	19.8% instruction cut; toy BSGS sometimes below rho	a weak reference creates false IC wins	current accounting-v2 rho, A/A, BSGS/kangaroo finite-size controls
Prime algebraic coverage	no complete native prime F4/F5 factor-base-constrained arm	generic root finding cannot answer the algebraic claim	implement and validate a prime algebraic reference arm
Evidence portability	mixed L0/L1 historical wall, one-curve studies	host effects and curve effects are confounded	L2 second host, at least 8 targets, 5 rounds, and 4 sizes

Bottleneck	Evidence now	Why it holds the frontier	Required next gate
GPU integration	verified 62-bit standalone result, no matched rho	capability result cannot enter the ECBench Pareto set	native adapter, same fixtures, matched reference, isolated clocks

The dominant pattern is that improvements are arriving after the point that currently controls total cost. Fast root extraction does not fix relation probability; fast linear algebra does not fix pair-table setup; fewer F6 XORs do not fix closure additions. The next useful optimization should be selected from the measured whole-pipeline phase share, not from the largest isolated microbenchmark speedup.

10 Curve-by-curve interpretation

10.1 Toy curves

Toy curves are valuable for exhaustive controls, replay, seed-law tests, solver parity, and detecting fixed overhead. They do not establish a security claim. At 16–26 prime bits, BSGS can beat rho in `bitop/v1`; at binary m=13, fixed setup can make the "strong" parallel rho variant worse than simple negation rho. The report therefore retains multiple generic controls.

10.2 Medium experimental curves

The repository's most informative generic boundary comes from ladders with enough sizes to reveal whether a fixed overhead is fading or a gap is growing. Prime pair/rootfinding costs move away from rho through 40/28 bits. Koblitz one-target IC remains several times rho through roughly 47 subgroup bits. The 62-bit GPU fixture is operationally larger but is not a matched frontier row.

10.3 Special transfers and covers

Isogeny, quartic-cover, GHS-cover, three-leads genus-2, and hyperelliptic paths must each retain the map, subgroup image, target transport, and inverse recovery cost. A cheaper DLP on the destination object helps only when those obligations are explicit and cheaper end to end. The GHS-weak family is a direct restricted problem class, not a free transfer from arbitrary curves.

10.4 Standard curves and ECC2K-130

No standard curve in the registry has a completed IC solve that changes its security boundary. The repository's ECC2K-130 material is a feasibility and solver program, not a demonstrated break. Claims about standard curves remain bounded by generic algorithms and external cryptanalytic knowledge, not by extrapolating toy ECBench fits.

11 Frozen fresh campaign

The prepared campaign contains nine independent L2 jobs. Each has eight targets, one warmup, six measured rounds with balanced arm order, replay-all, failure retention, and the three-point pricing band.

Spec	Domain/size	Principal candidate arms	Plan ID
`prime-p20.json`	prime 20	rho-v2, N45/N32 pair IC, rootfind m2/m3, BSGS, kangaroo, A/A	`ECS1h3320a6f156a5`
`prime-p24.json`	prime 24	rho-v2, N162/N115 pair IC, rootfind m2/m3, BSGS, kangaroo, A/A	`ECS1h8d2f84ff0bd1`
`prime-p28.json`	prime 28	rho-v2, N407/N288 pair IC, rootfind m2/m3, BSGS, kangaroo, A/A	`ECS1h280a82a43148`

Spec	Domain/size	Principal candidate arms	Plan ID
`prime-p32.json`	prime 32	rho-v2, N1024 pair IC, rootfind m2/m3, BSGS, kangaroo, A/A	`ECS1h483e365694a1`
`koblitz-n13.json`	Koblitz 13	strong/signed/negation rho, rootfind, inherited F4, F6, claw, A/A	`ECS1h5134645433fe`
`koblitz-n19.json`	Koblitz 19	same	`ECS1h729b865269f9`
`koblitz-n23.json`	Koblitz 23	same	`ECS1h1a916886cdf0`
`koblitz-n31.json`	Koblitz 31	same; work limits/timeouts retained	`ECS1h0ac2f6395870`
`solver-backends-k17.json`	binary solver cell 17	all 11 nested backends, strong rho, A/A	`ECS1h052a2276d7a5`

All specs pass `ecbench plan --json`. They are listed in execution/manifest.json. The locally built release binary has SHA-256 `91f5895637501c6de257bbaf3afc9cbc2c0f5e964964d3dc4fc3bf78a427d96b`.

Execution is intentionally blocked. This workspace has no AWS identity, no AWS CLI, no callable Iso Lab connector, and an invalid GitHub token. The only available pinned EKS image contains an older source revision, so using it would not measure this report's commit. The exact blocker is in execution/BLOCKER.md. Local wall timing is not silently substituted.

Before the first warmup, the executor must:

1. build the exact commit into the trusted static runtime image;
2. bind the immutable ECR digest and binary SHA-256 in the protocol;
3. validate and commit the frozen identity amendment;
4. run each size as a fresh job, without CRIU-restored timed rows;
5. upload raw records, host capsule, replay receipt, A/A analysis, and hashes;
6. retain every timeout, failure, and unattempted cell;
7. regenerate the Pareto tables and this report from the new records.

12 Conclusions and research priorities

The performance frontier is now explicit:

- **Generic ECC:** rho/BSGS/kangaroo remain the leaders; no generic IC crossing is admitted.
- **Restricted GHS-weak ECC:** plane-quartic IC is the leader and must remain labelled as a special-family result.
- **Prime engineering:** P20 $N = 45$ is the validated factor-base baseline, still roughly 8–10× rho.
- **Algebraic solvers:** F5 and F6 win component metrics; inherited F4 wins the tested common E2E lower-bound comparison; rho wins the ECDLP comparison.
- **SAT/WDSAT/FES:** implemented or historically explored, but not yet on one common, validated E2E frontier; WDSAT also carries a retained correctness failure in an old path.
- **Root finding:** valuable generic decomposition machinery, not evidence for an algebraic sub-square-root attack.
- **Rho:** instruction-level optimization is real and verified; isolated native wall promotion is still open.

The highest-value next work is therefore:

1. execute the already frozen L2 campaign on the exact source;
2. implement the missing prime factor-base-constrained algebraic arm;
3. attack relation yield and cold setup before further optimizing stages that are already a small share;
4. integrate the GPU relation engine with a matched ECBench reference;
5. preserve the GHS-weak win as a separate domain rather than using it to headline generic curves.

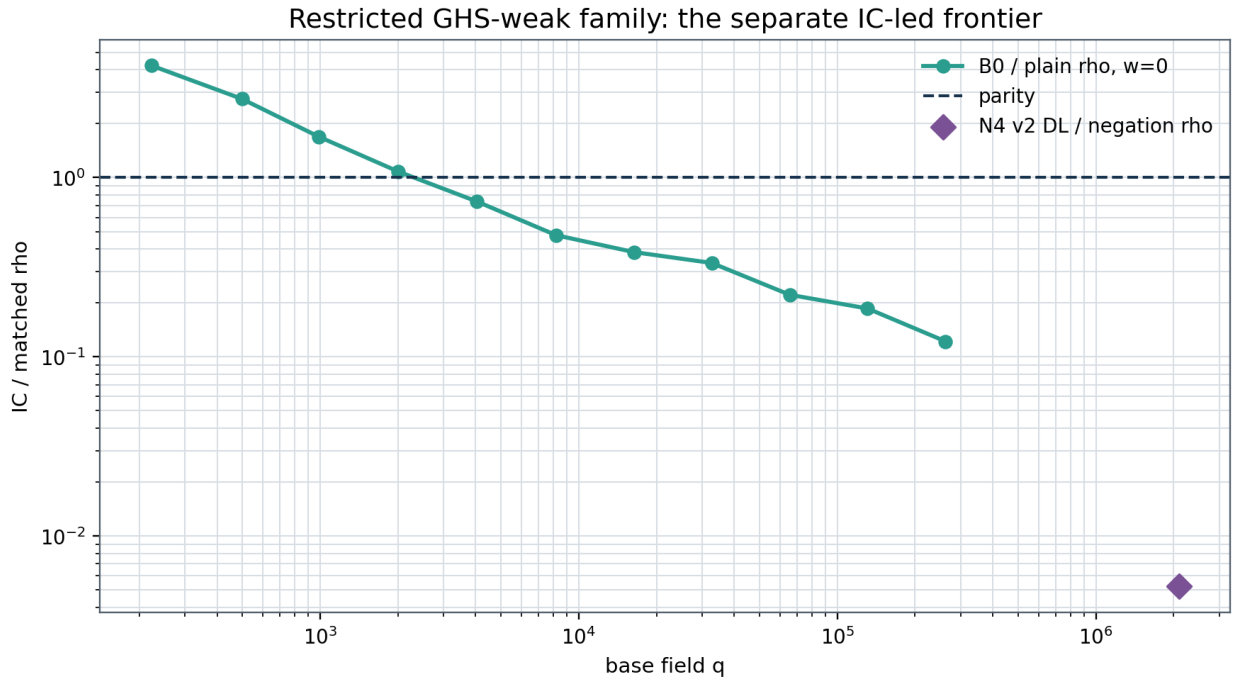


Figure 3: The GHS-weak curve family has its own, separate frontier.

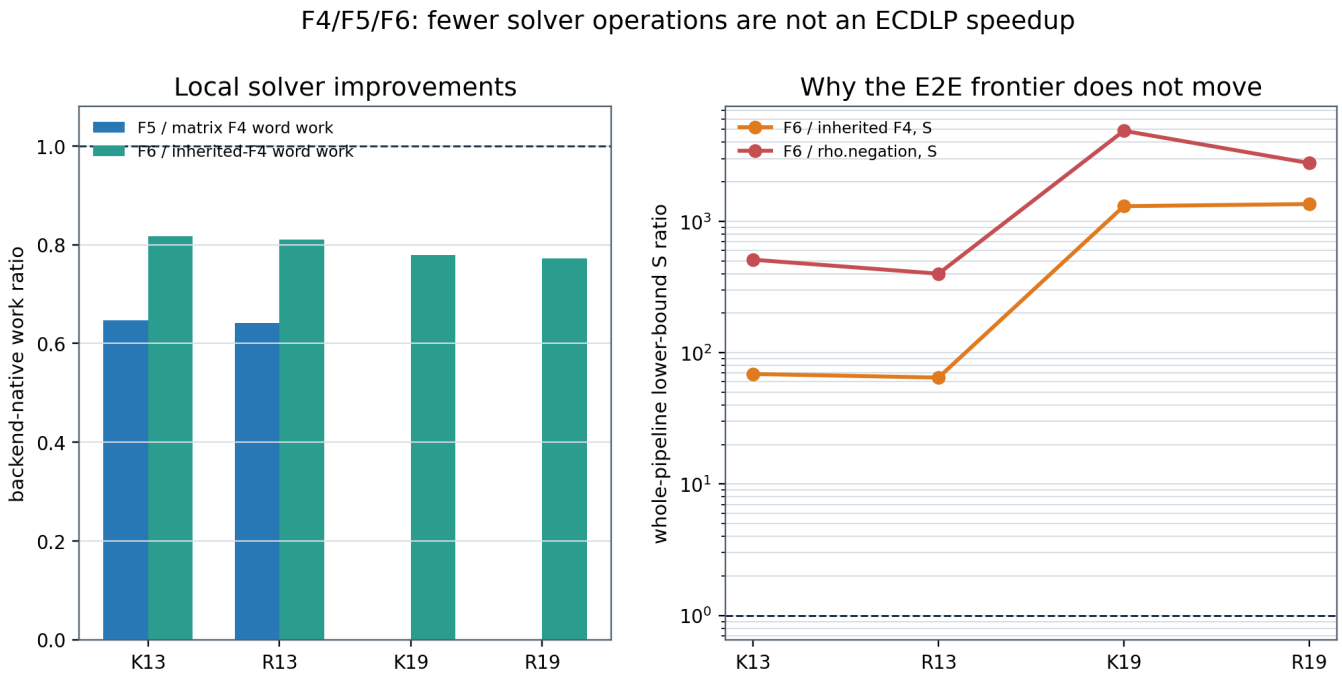


Figure 4: Backend-local savings and whole-pipeline cost move in opposite directions.

13 Reproduction and evidence index

The report is generated only from committed data and prior frozen evidence. Presentation data are in frontier_snapshot.json, and `render_figures.py` renders all four figures. Important sources are:

- canonical IC leaderboard;
- ECBench contract;
- backend-frontier contract;
- F4/F5/F6 result;
- rho optimization;
- prime factor-base second-host replay;
- GPU explicit-relations result;
- literature and prior-evidence ledger;
- live implementation coverage;
- campaign protocol.

Primary mathematical references used to set the boundary include Shoup's generic-group lower bound (EUROCRYPT 1997, DOI 10.1007/3-540-69053-0_18), Faugère's F4 (DOI 10.1016/S0022-4049(99)00005-5) and F5 (DOI 10.1145/780506.780516), Semaev's summation-polynomial work (IACR ePrint 2004/031), and the Faugère–Perret–Petit–Renault binary-field IC analysis (DOI 10.1007/978-3-642-29011-4_4).

No sentence in this report converts a missing measurement into a loss, a local counter into an E2E speedup, a special-family result into a generic break, or a draft EKS plan into fresh benchmark data.